

TVR Selection Software Report

1. Project Information

Date	1/1/2000
Project name	
Project address	
Country	Colombia
State	
City	Cali
Client name	
Client address	
Designed by	
Reference	New Project
Revision	
Altitude(m)	4915
Indoor DB temperature in cooling(°F)	73.5
Indoor WB temperature in cooling(°F)	59.7
Outdoor DB temperature in cooling(°F)	85
Outdoor WB temperature in cooling(°F)	70.7
Indoor DB temperature in heating(°F)	68
Indoor WB temperature in heating(°F)	57.2
Outdoor DB temperature in heating(°F)	73
Outdoor WB temperature in heating(°F)	67.8

2. Overall Material List

2.1 Equipment List

Model	Quantity	Description
4TVY0096H8000AA	1	TVR Pro CO (220V)
4TVC0048EF000AA	2	Four-way Cassette
TRDK112HP	1	Branch joint
TCONTRMUT05B	2	Remote controller TVR Ultra

2.2 Field Providing List

2.2.1 Refrigerant Piping Materials

Model	Quantity	Unit	Description
Φ3/8	9	m	Copper pipe
Φ5/8	6	m	Copper pipe
Φ7/8	3	m	Copper pipe
Insulation casing for piping			All refrigerant piping and branch joints should be completely insulated.

Recommended insulation casing thickness:

Piping size	Thickness	
	Humidity<80%RH	Humidity≥80%RH
Φ6.35~Φ38.1mm	≥15mm	≥20mm
Φ41.3~Φ38.1mm	≥20mm	≥25mm

2.2.2 Refrigerant charge

System name	Model	Quantity	Unit	Description
System1	R410A	0.54	kg	Extra Refrigerant Added

2.2.3 Electrical cables

Type	Size	Length
Power supply cable	Select based on MCA of each unit	According to the actual system design
Communication cable	PQE:0.75mm ² three-core shielded cable	According to the actual system design

3. Overall Electrical Characteristics

Model	Quantity	Power supply	MCA(A)	MFA(A)
4TVY0096H8000AA	1	220V-3ph-60Hz	42.80	50
4TVC0048EF000AA	2	220-240V-50/60Hz	1.20	15

Notes:

1. MCA: Minimum Circuit Amps. MCA is used to select wire size.The value in above table is for one unit.

2. MFA: Maximum Fuse Amps. MFA is used to select overcurrent circuit breakers and residual-current circuit breakers.The value in above table is for one unit.

4. System1

4.1 BOM List (System1)

Model	Quantity	Unit	Description
4TVY0096H8000AA	1		TVR Pro CO (220V)
4TVC0048EF000AA	2		Four-way Cassette
TRDK112HP	1		Branch joint
TCONTRMUT05B	2		Remote controller TVR Ultra
R410A	0.54	kg	Extra Refrigerant Added
Φ3/8	9	m	Copper pipe
Φ5/8	6	m	Copper pipe
Φ7/8	3	m	Copper pipe

4.2 Indoor Unit Details (System1)

4.2.1 Indoor Unit Details Table

IDU Name	Model	Weight(kg)	Dimension(WxHxD)(mm)	Power supply	MCA(A)	MFA(A)
IDU1	4TVC0048EF000AA	30.7	840*300*840	220-240V-50/60Hz	1.2	15
IDU1	4TVC0048EF000AA	30.7	840*300*840	220-240V-50/60Hz	1.2	15

IDU Name	Model	Tmp-C(°F)	RTC(Btu/h)	ATC(Btu/h)	RSC(Btu/h)	ASC(Btu/h)	PI-C(W)	Tmp-H(°F)	RHC(Btu/h)	AHC(Btu/h)	PI-H(W)
IDU1	4TVC0048EF000AA	73.5/59.7		39,861		36,690	170	68		49,228	170
IDU1	4TVC0048EF000AA	73.5/59.7		39,861		36,690	170	68		49,228	170

IDU Name	Model	Airflow(CFM)	Sound-Pr dB(A)	ESP(in.w.g.)
IDU1	4TVC0048EF000AA	1057.35[SSH]	45[SSH]	0
IDU1	4TVC0048EF000AA	1057.35[SSH]	45[SSH]	0

IDU Name	Model	Piping Length to 1st Y Joint(m)
IDU1	4TVC0048EF000AA	3.00
IDU1	4TVC0048EF000AA	3.00

4.2.2 Table of Abbreviations

Abbreviation code	Description
Tmp-C	Indoor temperature in cooling (Dry bulb temp. / Wet bulb temp. / RH)
RTC	Required total cooling capacity
ATC	Available total cooling capacity
RSC	Required sensible cooling capacity
ASC	Available sensible cooling capacity
Tmp-H	Indoor temperature in heating (Dry bulb temp.)
RHC	Required heating capacity
AHC	Available heating capacity
Tdis-H	Indoor unit discharge air temperature in heating
Airflow	Indoor unit airflow (High/Medium/Low)
ESP	External static pressure
Sound-Pr	Sound pressure level (High/Medium/Low)
Sound-Po	Sound power level (High/Medium/Low)
MCA	Minimum Circuit Amps
MFA	Maximum Fuse Amps
PI-C	Power input in cooling
PI-H	Power input in heating
Power supply	Power supply
Dimension(WxHxD)	Net Dimension (WxHxD) mm
Weight	Weight

4.3 Outdoor Unit Details (System1)

4.3.1 Outdoor Unit Details Table

Model	4TVY0096H8000AA
Module	4TVY0096H8000AA
Tmp-C	85
RTC	Btu/h

ATC	Btu/h	70,162
PI-C	kW	3.81
EER		5.39
Tmp-H	°F/°F	73/67.8
RHC	Btu/h	
AHC	Btu/h	0
PI-H	kW	-0.00
COP		
CR		100.0
Airflow	CFM	6109.13
Sound-Pr		58
Sound-Po		
Bas-Refr	kg	8.00
Ex-Refr(ODU)	kg	0.00
Ex-Refr(Piping)	kg	0.54
TCO2 eq.		17.83
MCA	A	42.8
MFA	A	50
Power supply	V/ph/Hz	220V-3ph-60Hz
Dimension (WxHxD)	mm	960*1615*765
Weight	kg	193

4.3.2 Table of Abbreviations

Abbreviation code	Description
Tmp-C	Outdoor conditions in cooling (Dry bulb temp.)
RTC	Required cooling capacity
ATC	Available cooling capacity
PI-C	Power input in cooling
EER	EER
Tmp-H	Indoor conditions in heating (Dry bulb temp. / Wet bulb temp. / RH)
RHC	Required heating capacity
AHC	Available heating capacity
PI-H	Power input in heating
COP	COP
CR	Combination ratio
Airflow	Outdoor unit airflow
Sound-Pr	Sound pressure level
Sound-Po	Sound power level
Bas-Refr	Standard factory refrigerant charge
Ex-Refr(ODU)	Extra refrigerant charge for outdoor unit
Ex-Refr(Piping)	Extra refrigerant charge for liquid piping
TCO2 eq.	Tonnes of CO2 equivalent
MCA	Minimum Circuit Amps
MFA	Maximum Fuse Amps
Power supply	Power supply
Dimension (WxHxD)	Net Dimension (WxHxD) mm
Weight	Weight

4.4 Piping Limitations (System1)

4.4.1 Piping Limitations

Item	Capability	Actual Value
Total piping length	1000.00(m)	9.50(m)
Longest actual length	175.00(m)	6.00(m)
Longest equivalent length	200.00(m)	6.50(m)
Longest equivalent length after first branch	90.00(m)	3.00(m)
Indoor unit to nearest branch length	40.00(m)	3.00(m)
Length difference between longest and shortest distance to indoor units	40.00(m)	0.00(m)
Height difference between indoor and outdoor unit(ODU up)	90.00(m)	0.00(m)
Height difference between indoor and outdoor unit(ODU down)	110.00(m)	0.00(m)
Height difference between indoor units	30.00(m)	0.00(m)
Combination ratio	50-130%	100.00%
IDU quantity	16	2

4.4.2 Correction Factors

Item	Correction factor
Altitude (indoor unit)	0.902
Altitude (outdoor unit)	0.961
Piping (cooling)	1.000
Piping (heating)	1.000
Defrost (heating)	1.000

4.4.3 Piping Details Table

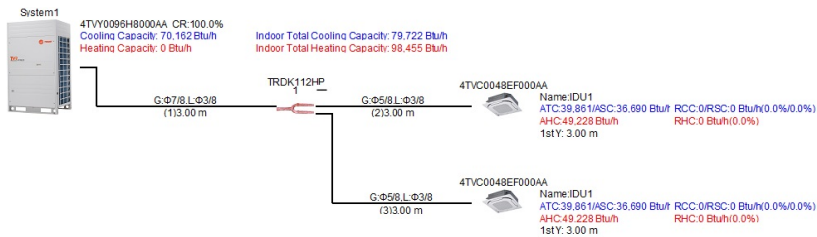
No.	Length(m)	Piping diameter
(1)	3.00	Φ7/8/Φ3/8
(2)	3.00	Φ5/8/Φ3/8
(3)	3.00	Φ5/8/Φ3/8

4.4.4 Branch Joints Details Table

No.	Load(Btu/h)	Model
(1)	95,540	TRDK112HP

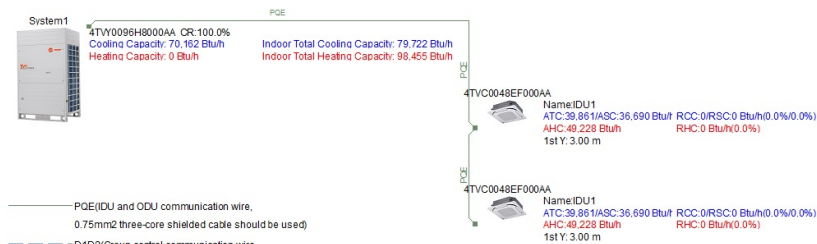
4.4.5 Reducer Details Table

4.5 Piping Diagrams (System1)



The drawing may differ from actual conditions due to limitations of the software. Please confirm before installation according to the installation manual.

4.6 Wiring Diagrams (System1)



- PQE(IDU and ODU communication wire, 0.75mm² three-core shielded cable should be used)
- D1D2(Group control communication wire, 0.75mm² two-core shielded cable should be used)
- ABCD(E)Infrared communication wire, 0.5mm² four-core shielded cable should be used
- X1X2 (Power line communication wire, 0.75mm² two-core shielded cable should be used)
- D1D2+X1X2(Group control communication and power wire 0.75mm² two-core shielded cable +0.75mm² two-core shielded cable should be used)
- XYE (ODU and ODU communication wire, 0.75mm² three-core shielded cable should be used)
- Power wire

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5. Centralized Control Solution

5.1 Centralized Controller List

The centralized control system of this project is full output regardless of whether the system is selected.